

ORIGINAL ARTICLE

Calcium Channel Blockade Versus β -Blockade for Hypertension in Heart Failure With Preserved Ejection Fraction: A Randomized Crossover Trial

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BACKGROUND: Hypertension is present in 90% of individuals with heart failure with preserved ejection fraction (HFpEF) and is a major modifiable risk factor for the development of HFpEF. However, randomized controlled trial evidence for hypertension management in HFpEF is limited.

METHODS: In a double-blind, randomized, crossover trial, we studied the effect of amlodipine 5 to 10 mg versus metoprolol succinate 100 to 200 mg (doses previously demonstrated to have comparable antihypertensive efficacy) for 4 weeks among adults with HFpEF and hypertension, without contraindications to initiating or withholding either drug. The primary outcome was the difference in mean home systolic blood pressure (BP) during the final week of each treatment.

RESULTS: The mean age of the 50 enrolled participants was 72 ± 9 years, 34 (68%) were female, 33 (66%) were of Black race, mean blood pressure was $144 \pm 15/78 \pm 9$ mmHg, and 23 (46%) were receiving β -blockers before enrollment. Compared with metoprolol, systolic BP was 4 (95% CI, -7 to -1 ; $P=0.017$) mmHg lower with amlodipine. In addition, peak oxygen uptake during exercise was 1.2 (95% CI, 0.3–2.0; $P=0.008$) mL/min per kg higher, physical activity was 0.1 (95% CI, 0.01–0.1; $P=0.019$) metabolic equivalents of task/d higher, and NT-proBNP (N-terminal pro-B-type natriuretic peptide) was 200 (95% CI, -291 to -109 ; $P<0.0001$) pg/mL lower with amlodipine versus metoprolol. There was no significant difference in septal E/e', myocardial strain, or systemic vasodilatory reserve. The frequency and severity of adverse events were similar across treatments.

CONCLUSIONS: Our findings support the use of dihydropyridine calcium channel blockers as a preferred alternative to β -blockers for the management of hypertension in HFpEF. (**Hypertension**. 2026;83:2142–2152. DOI: 10.1161/HYPERTENSIONAHA.126.26801.) • **Supplement Material**.

REGISTRATION: URL: <https://www.clinicaltrials.gov>; Unique identifier: NCT04434664.

Key Words: antihypertensive agents ■ blood pressure ■ exercise tolerance ■ heart failure ■ hypertension

Heart failure (HF) affects over 6.7 million adults in the United States and is a major source of morbidity, mortality, and impaired quality of life.¹ Over half of individuals with HF have a preserved left ventricular (LV) ejection fraction (EF), termed HF with preserved EF (HFpEF),^{2,3} and $\approx 90\%$ of these individuals have a diagnosis of hypertension.⁴ Hypertension is a crucial target

for therapeutic interventions in patients with HFpEF due to its high prevalence and important pathophysiologic role in disease development.⁵ However, data from randomized controlled trials studying hypertension management in HFpEF are limited.

Dihydropyridine calcium channel blockers (CCBs) are recommended as first-line agents for the management

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NOVELTY AND RELEVANCE

What Is New?

This is the first randomized, double-blind, crossover trial directly comparing a dihydropyridine calcium channel blocker with a β -blocker for hypertension management in heart failure with preserved ejection fraction. Amlodipine lowered home and central systolic blood pressure more than metoprolol. Amlodipine improved exercise capacity, physical activity, and NT-proBNP (N-terminal pro-B-type natriuretic peptide) compared with metoprolol.

What Is Relevant?

Hypertension is nearly universal in heart failure with preserved ejection fraction, yet evidence guiding anti-hypertensive drug choice is limited. β -blockers are commonly prescribed despite lack of proven benefit in heart failure with preserved ejection fraction without another indication.

Clinical/Pathophysiological Implications?

Dihydropyridine calcium channel blockers may be preferred over β -blockers for treating hypertension in heart failure with preserved ejection fraction to improve blood pressure control and functional capacity.

Nonstandard Abbreviations and Acronyms

BP	blood pressure
CCB	calcium channel blocker
COPD	chronic obstructive pulmonary disease
DELIVER	Dapagliflozin Evaluation to Improve the Lives of Patients With Preserved Ejection Fraction Heart Failure
EF	ejection fraction
HF	heart failure
HFpEF	heart failure with preserved ejection fraction
LV	left ventricular
NT-proBNP	N-terminal pro-B-type natriuretic peptide
TOPCAT	Treatment of Preserved Cardiac Function Heart Failure With an Aldosterone Antagonist trial

of hypertension in the general population. Although β -blockers are currently not recommended for first-line antihypertensive therapy in the general population,^{6,7} they reduce the risk of mortality in patients with HF with reduced EF.^{8,9} Yet, no consistent cardiac or mortality benefit has been demonstrated in those with HFpEF in the absence of arrhythmias.^{10–12} Nonetheless, HF guidelines historically recommended β -blockers, and not CCBs, for the management of hypertension in patients with HFpEF, based on extrapolation from HF with reduced EF data.^{8–10} While the most recent HF guidelines removed the recommendation for β -blockers,¹³ these agents are used much more commonly than CCBs in HFpEF. In the TOPCAT trial (Treatment of Preserved Cardiac Function Heart Failure With an Aldosterone Antagonist), 78% of

participants were on β -blockers at baseline, while only 38% were on CCBs.¹⁴ Similarly, in the DELIVER trial (Dapagliflozin Evaluation to Improve the Lives of Patients With Preserved Ejection Fraction Heart Failure), 82% of participants were on β -blockers.¹⁵

In the general population, CCBs have beneficial effects related to vasodilation and arterial function beyond their blood pressure (BP)-lowering effect.^{16,17} These vasodilatory properties may reduce LV afterload, improve ventricular-vascular coupling, and improve oxygen utilization in HFpEF. However, nonselective vasodilation at rest may also reduce the exercise-induced vasodilatory reserve in skeletal muscle and impair blood flow redistribution. Given the multiple potential mechanisms by which CCBs and β -blockers may affect BP and, particularly, the determinants of aerobic capacity in HFpEF, we performed a double-blind, randomized crossover trial to evaluate the comparative effectiveness of CCBs versus β -blockers on home systolic BP, exercise tolerance, and LV diastolic function in this patient population. We hypothesized that CCBs would be more effective for BP lowering than β -blockers and that these medications exert different effects on exercise/aerobic capacity and its physiological determinants.

METHODS

Data Availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Study Design and Oversight

We conducted a double-blind, randomized, crossover trial at the University of Pennsylvania. Participants were enrolled from December 1, 2021, to December 3, 2024. The trial design was approved by the institutional review board at the University of Pennsylvania. An independent data safety monitoring board

provided oversight of the trial. Details of the rationale and trial design are included in the [Supplemental Material](#). All participants provided written informed consent. Participants were recruited from outpatient HF clinics after electronic health record review.

Participants

See the [Supplemental Material](#) for detailed inclusion and exclusion criteria. Briefly, participants were required to be 18 to 90 years of age with a diagnosis of hypertension on stable anti-hypertensive therapy and HFpEF based on clinical assessment, LVEF >50%, and signs of elevated LV filling pressures on echocardiogram or right heart catheterization. Participants were excluded if they had a seated office systolic BP <100 mmHg, ≥180 mmHg if not taking a CCB or β -blocker at the time of screening, or ≥160 mmHg if taking a CCB or β -blocker (which would subsequently need to be held for the trial); resting heart rate <50 or >100 bpm; orthostatic hypotension; or any contraindication to initiating or withholding CCBs or β -blockers.

Randomization and Masking

Eligible participants were randomly assigned to initial therapy with amlodipine or metoprolol succinate. Amlodipine 5 mg tablets and metoprolol succinate 100 mg tablets were placed into identical oral gelatin capsules. Investigators, treating clinicians, and participants were blinded to the treatment order. Randomization lists were generated using a standard random number generator. The treatment order assignments and the associated study-specific participant identification numbers were concealed and maintained by the University of Pennsylvania Investigational Drug Service using a centralized, pharmacy-controlled system. After each participant's eligibility was confirmed, a designated pharmacist determined and dispensed the treatment assignment independently of the investigators, such that participants and investigators remained blinded.

Interventions

For amlodipine, participants were initiated on 5 mg daily (ie, 1 capsule identical to metoprolol) and titrated up to 10 mg (ie, 2 capsules) daily after 1 week. For metoprolol, participants were initiated on 100 mg daily (ie, 1 capsule identical to amlodipine) and titrated up to 200 mg (ie, 2 capsules) daily. For both therapies, criteria for up-titration were home systolic BP ≥130 mmHg and heart rate ≥50 bpm after the first week of use. Participants received each medication for 4 weeks, with a 1-week washout period before initiating each medication. Medication adherence was assessed via pill count at each end point visit.

Outcomes

The primary end point was the difference in mean home systolic BP during the final week of each intervention. All participants performed home BP monitoring using the same device model (Microlife BP 3MX1-4 WatchBP Home N,¹⁸ Microlife Corporation, Taiwan, ROC) and were carefully instructed on appropriate technique,¹⁹ including 3 successive measurements in the morning and evening over the 7-day period. A minimum of 3 days of twice-daily BP readings were required to be considered valid.²⁰

Key secondary end points were differences in measures of (1) BP, including office systolic BP and home and office diastolic BP (determined by the average of 3 automated readings using standardized technique, performed by a trained member of the research team²¹); (2) exercise tolerance, including peak oxygen uptake (peak VO_2); (3) physical activity, including metabolic equivalents of tasks on actigraphy; (4) LV function, including NT-proBNP (N-terminal pro-B-type natriuretic peptide), septal E/e', and global longitudinal myocardial strain; (5) quality of life, determined by the Kansas City Cardiomyopathy Questionnaire overall summary score; (6) exercise systemic vasodilatory reserve; and (7) arterial hemodynamic parameters, including carotid-femoral pulse wave velocity and central systolic BP, and metrics of arterial load derived from pressure-flow analyses as previously described.²² The study end points are described in detail in the [Supplemental Methods](#).

Statistical Analyses

Using a conservative SD of 10.6 mmHg from a prior study of amlodipine versus metoprolol in individuals with hypertension,²³ we had 90% power at an alpha of 0.05 to detect a within-subject difference in mean daytime systolic BP of 5 mmHg between the different therapies (primary outcome), including prespecified subgroup analysis of the primary outcome by sex.^{24–26}

Analyses were performed on a modified intention-to-treat basis using the total number of randomized participants with available end point data. Descriptive estimates of baseline characteristics by initial treatment group present estimates of central tendency (median, mean) and measures of variability (25th–75th percentile, SD). For the primary analyses, the effects of amlodipine versus metoprolol on the primary and secondary end points were assessed using the paired *t* test and the nonparametric Wilcoxon signed-rank test on the difference between the paired within-subject outcome measures across the 2 therapies. These paired methods estimate the within-participant treatment contrast among participants with complete paired outcome data. Secondary analyses were performed using linear mixed-effects models accounting for period, sequence, and a random subject effect nested within sequence, which allowed for inclusion of data from dropouts,^{27–29} a priori adjustment for age, body mass index, estimated glomerular filtration rate, and number of BP medications,^{30,31} with further adjustment for any imbalanced covariates. The mixed-effects models also estimate the within-participant treatment contrast while additionally accounting for period and sequence effects and allowing inclusion of participants with incomplete outcome data. Restricted maximum likelihood estimation was used, and an appropriate covariance matrix was specified.

In prespecified subgroup analyses, results were stratified and interactions were assessed by several prespecified potential effect modifiers, including sex, race, diabetes status, chronotropic incompetence, chronic obstructive pulmonary disease (COPD) status, New York Heart Association HF class, obesity status, and chronic kidney disease status. For the stratified analyses, *P* values were corrected with Bonferroni adjustment for multiple comparisons.

Analyses were performed using SAS, version 9.4 M8 (SAS Institute, Inc, Cary, NC) and R, version 4.3.2 (R Foundation for Statistical Computing, Vienna, Austria).

RESULTS

Study Participants

A total of 50 participants were randomized, 25 to initial therapy with amlodipine and 25 to initial therapy with metoprolol between December 1, 2021, and December 3, 2024 (see Figure 1 and Figure S1 for additional details regarding exclusions). There were 4 participants who withdrew before receiving treatment, 2 from each randomization arm. Two additional participants withdrew after starting treatment, one of whom withdrew after providing complete primary end point data, allowing them to be included in the primary end point analysis. Accordingly, 45 participants were included in the primary analysis.

The mean age of the 50 enrolled participants was 72 ± 9 years, 34 (68%) were female, 33 (66%) were Black race, mean office BP was $144 \pm 15/78 \pm 9$ mm Hg, median body mass index was 35 (31, 41) kg/m², 16 (32%) had COPD, median NT-proBNP was 97 (74, 177) pg/mL, mean LVEF was $63 \pm 7\%$, and mean resting E/e' was 14 ± 4 . Baseline characteristics were generally similar across the 2 treatment groups (see Table 1 and Table S1 for more detailed baseline characteristics), although those who received amlodipine versus metoprolol first were less likely to have COPD (16% versus 48%) and less likely to be female (52% versus 84%).

Primary End Point

In the primary analyses, among the 45 participants with complete paired data, mean systolic home BP was 4 (95% CI, -7 to -1) mm Hg lower during treatment with amlodipine compared with metoprolol (131 versus 135 mm Hg; $P=0.017$; Table 2; Figure 2A). The results were consistent in mixed-effects models adjusted for period, sequence, and prespecified (age, body mass index, estimated glomerular filtration rate, number of BP medications) and imbalanced (sex and COPD status) covariates (least squares mean difference, -4 mm Hg; 131 versus 134 mm Hg [95% CI, -7 to -1]; $P=0.023$; Table 3).

Secondary End Points

In unadjusted analyses, during treatment with amlodipine versus metoprolol, peak $\dot{V}O_2$ was 1.2 mL/min per kg higher (12 versus 11 mL/min per kg [95% CI, 0.3–2.0]; $P=0.008$; Figure 2B), physical activity was 0.1 metabolic equivalents of task/d higher (1.5 versus 1.4 metabolic equivalents of task/d [95% CI, 0.01–0.1]; $P=0.019$), NT-proBNP was 200 pg/mL (133 versus 344 pg/mL [95% CI, -291 to -109]; $P<0.0001$) lower (Figure 2C), and central systolic BP was 8 mm Hg (124 versus 129 mm Hg [95% CI, -14 to -2]; $P=0.013$) lower. While most other secondary end points were directionally better with amlodipine versus metoprolol, there was no significant

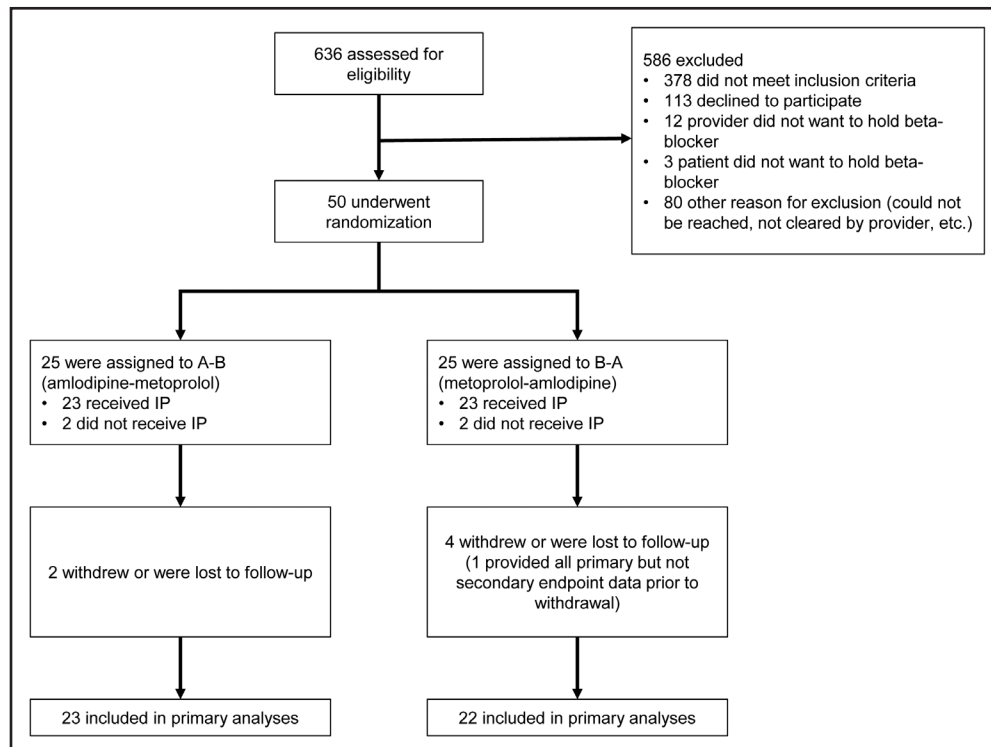


Figure 1. Participant enrollment, randomization, and follow-up.

Table 1. Baseline Demographic and Clinical Characteristics by Randomization Arm

Parameter	All participants, N=50	Amlodipine first (A-B), N=25	Metoprolol First (B-A), N=25
Age, y	72 (9)	72 (10)	73 (9)
Female sex, N (%)	34 (68)	13 (52)	21 (84)
Black race, N (%)	33 (66)	15 (60)	18 (72)
Office systolic BP, mm Hg	144 (15)	145 (15)	142 (14)
Office diastolic BP, mm Hg	78 (9)	79 (11)	77 (7)
Home systolic BP, mm Hg	141 (17)	141 (18)	140 (16)
Home diastolic BP, mm Hg	77 (10)	78 (10)	77 (11)
Heart rate, bpm	71 (12)	70 (11)	72 (13)
BMI, kg/m ²	35 (31–41)	35 (30–41)	35 (32–42)
Diabetes, N (%)	18 (36)	6 (24)	12 (48)
Coronary artery disease, N (%)	5 (10)	3 (12)	2 (8)
Atrial fibrillation/atrial flutter (paroxysmal or resolved), N (%)	15 (30)	5 (20)	10 (40)
Hyperlipidemia, N (%)	43 (86)	22 (88)	21 (84)
COPD, N (%)	16 (32)	4 (16)	12 (48)
New York Heart Association Class, N (%)			
Class II	41 (82)	20 (80)	21 (84)
Class III	9 (18)	5 (20)	4 (16)
CCB use, N (%)	32 (64.0)	16 (64.0)	16 (64.0)
β-Blocker use, N (%)	23 (46.0)	13 (52.0)	10 (40.0)
BP medications, number	3 (2–3)	3 (2–3)	3 (2–3)
eGFR, mL/min per 1.73 m ²	70 (19)	67 (18)	73 (20)
Hemoglobin, g/dL	13 (2)	13 (2)	13 (1)
NT-proBNP, pg/mL	97 (74–177)	97 (82–168)	100 (69–177)
LVEF, %	62.76 (7.32)	61.96 (7.00)	63.56 (7.68)
Septal E/e' ratio	14 (4)	13 (4)	14 (4)
Carotid-femoral pulse wave velocity, m/s	10 (7–13)	10 (7–13)	9 (7–13)
Global longitudinal myocardial strain, %	−19 (3)	−18 (3)	−19 (3)
KCCQ, score	68 (23)	68 (25)	68 (22)

Continuous variables are expressed as mean (SD) or median (25th–75th percentile). BMI indicates body mass index; BP, blood pressure; CCB, calcium channel blocker; COPD, chronic obstructive pulmonary disease; eGFR, estimated glomerular filtration rate; KCCQ, Kansas City Cardiomyopathy Questionnaire; LVEF, left ventricular ejection fraction; and NT-proBNP, N-terminal pro-B-type natriuretic peptide.

difference in office BPs, home diastolic BP, systemic vasodilatory reserve, or other measures of LV diastolic function and arterial hemodynamics (Table 2; Table S2; Figure S2). Medication adherence was high and did not differ meaningfully between amlodipine and metoprolol (Table S2).

There was a significant interaction between period and sequence for peak VO_2 (Table S3). After adjustment for this and covariates in mixed-effects models, results were similar to the unadjusted analyses (Table 3; Table S4).

Prespecified Subgroup Analyses

There was no effect modification by sex, race, diabetes status, chronotropic incompetence, COPD status, New York Heart Association HF class, obesity, or chronic kidney disease status with regard to home systolic BP, peak VO_2 , and NT-proBNP (see Figure S3 for

Bonferroni-adjusted stratified *P* values, and Table S5 for treatment-by-subgroup interaction *P* values).

Adverse Events

The severity and frequency of adverse events were similar across treatments (Table S6), with higher rates of high BP and bradycardia noted during treatment with metoprolol (Table S7).

DISCUSSION

Among individuals with hypertension and HFpEF, amlodipine reduced home systolic BP and central systolic BP, improved peak oxygen uptake during exercise, increased ambulatory activity, and lowered NT-proBNP compared with metoprolol. We also observed improvements in several other measures of BP, ambulatory activity, LV function, and arterial load during treatment with amlodipine

Table 2. Effect of Amlodipine Versus Metoprolol Succinate on the Primary and Selected Secondary End Points

	Mean (SD)		Estimated difference between amlodipine and metoprolol (95% CI)	P value
	Amlodipine	Metoprolol		
Primary outcome				
Home systolic BP, mm Hg	131 (12)	135 (17)	−4 (−7 to -1)	0.017
Secondary outcomes				
Home diastolic BP, mm Hg	74 (8)	74 (8)	−0.6 (−2 to 1)	0.36
Office systolic BP, mm Hg	133 (33)	136 (21)	−4 (−15 to 7)	0.49
Office diastolic BP, mm Hg	72 (11)	73 (15)	−1 (−5 to 3)	0.59
Peak oxygen uptake, ml/min per kg	12 (4)	11 (3)	1.2 (0.3 to 2.0)	0.008
Physical activity, average METs/d	1.5 (0.2)	1.4 (0.1)	0.1 (0.01 to 0.1)	0.019
Moderate- to vigorous-intensity physical activity, min/d	72 (46)	60 (32)	10 (−0.2 to 20)	0.05
Systemic vasodilatory reserve, %	20 (14)	25 (19)	−5 (−13 to 4)	0.25
KCCQ, score	67 (22)	68 (24)	−2 (−7 to 3)	0.43
Septal E/e' ratio	14 (5)	15 (5)	−1 (−2 to 0.2)	0.13
LVEF, %	63 (5)	63 (6)	0 (−2 to 2)	0.9
Global longitudinal myocardial strain, %	−19 (3)	−20 (2)	0.5 (−0.1 to 1)	0.09
NT-proBNP, pg/mL	133 (194)	344 (424)	−200 (−291 to −109)	<0.0001
Carotid-femoral pulse wave velocity, m/s	11 (5)	10 (5)	0.4 (−1 to 2)	0.59
Central systolic BP, mm Hg	124 (16)	129 (19)	−8 (−14 to −2)	0.013
Augmentation index, %	139 (27)	151 (31)b	−11 (−23 to 2)	0.1

Analyses used the (unadjusted) paired *t* test or Wilcoxon sign-rank test and were restricted to individuals with complete data for each end point; all available end point data are presented in the first 2 columns. The third column presents paired differences among those with complete data for both end points. Moderate- to vigorous-intensity physical activity was defined as activities ≥ 3 METs. BP indicates blood pressure; KCCQ, Kansas City Cardiomyopathy Questionnaire; LVEF, left ventricular ejection fraction; MET, metabolic equivalents of task; and NT-proBNP, N-terminal pro-B-type natriuretic peptide.

versus metoprolol that were not statistically significant. There were no meaningful differences in the relationship between amlodipine versus metoprolol and the primary and key secondary outcomes across multiple subgroups, including sex, chronotropic incompetence, and New York Heart Association class. Overall rates and severity of adverse events were similar across treatments, with individuals more likely to experience headaches, dizziness, and bradycardia during treatment with metoprolol versus amlodipine.

The current trial was designed in the context of clinical uncertainty about whether dihydropyridine CCBs or β -blockers offer greater benefit for patients with hypertension and HFpEF. The specific agents and dosing were selected based on a prior double-blind, crossover trial that randomized subjects with a diagnosis of hypertension, who were otherwise healthy, to amlodipine 5 to 10 mg daily versus metoprolol succinate 100 to 200 mg daily.²³ That trial demonstrated no significant difference in ambulatory BP between arms. In addition, metoprolol is the most commonly prescribed β -blocker in the United States, accounting for 59% of β -blocker prescriptions. Nonetheless, our findings may not generalize to vasodilating β -blockers, and comparative trials against third-generation β -blockers are warranted. Importantly, data from uncomplicated hypertension cannot be readily extrapolated to HFpEF, given the complex effects of

antihypertensive agents (including vasoactive, chronotropic, inotropic, and lusitropic effects) on key determinants of aerobic capacity in HFpEF. Although patients with HFpEF are more likely to be prescribed β -blockers than CCBs,^{14,15} data supporting this is not available in the HFpEF literature. Meta-analyses of data from 3 randomized trials suggest that β -blockers may reduce cardiovascular mortality in HFpEF compared with placebo or usual care (hazard ratio, 0.75 [95% CI, 0.60–0.94]),¹¹ with no effect on HF hospitalization or quality of life.^{10,11} However, these trials were limited in quality, and used definitions for HFpEF that are not consistent with current guidelines (HFpEF was defined using an LVEF threshold of $\geq 35\%$ ³² or $\geq 40\%$,^{33,34} compared with the consensus definition of LVEF $\geq 50\%$,⁹ thus likely representing HF with mid-range EF rather than HFpEF). Observational studies have shown conflicting results on the effects of β -blockers on HF-related outcomes in HFpEF, with some suggesting protection against HF hospitalizations and cardiac death¹⁵ and others suggesting increased risk.^{35,36} Of note, randomized controlled trials comparing the effects of specific antihypertensive agents on BP, aerobic capacity, and cardiovascular function in HFpEF are lacking.

Data from a prior trial suggest benefits from withdrawing β -blockers in certain individuals with HFpEF. Palau et al³⁷ conducted a crossover mechanistic trial evaluating

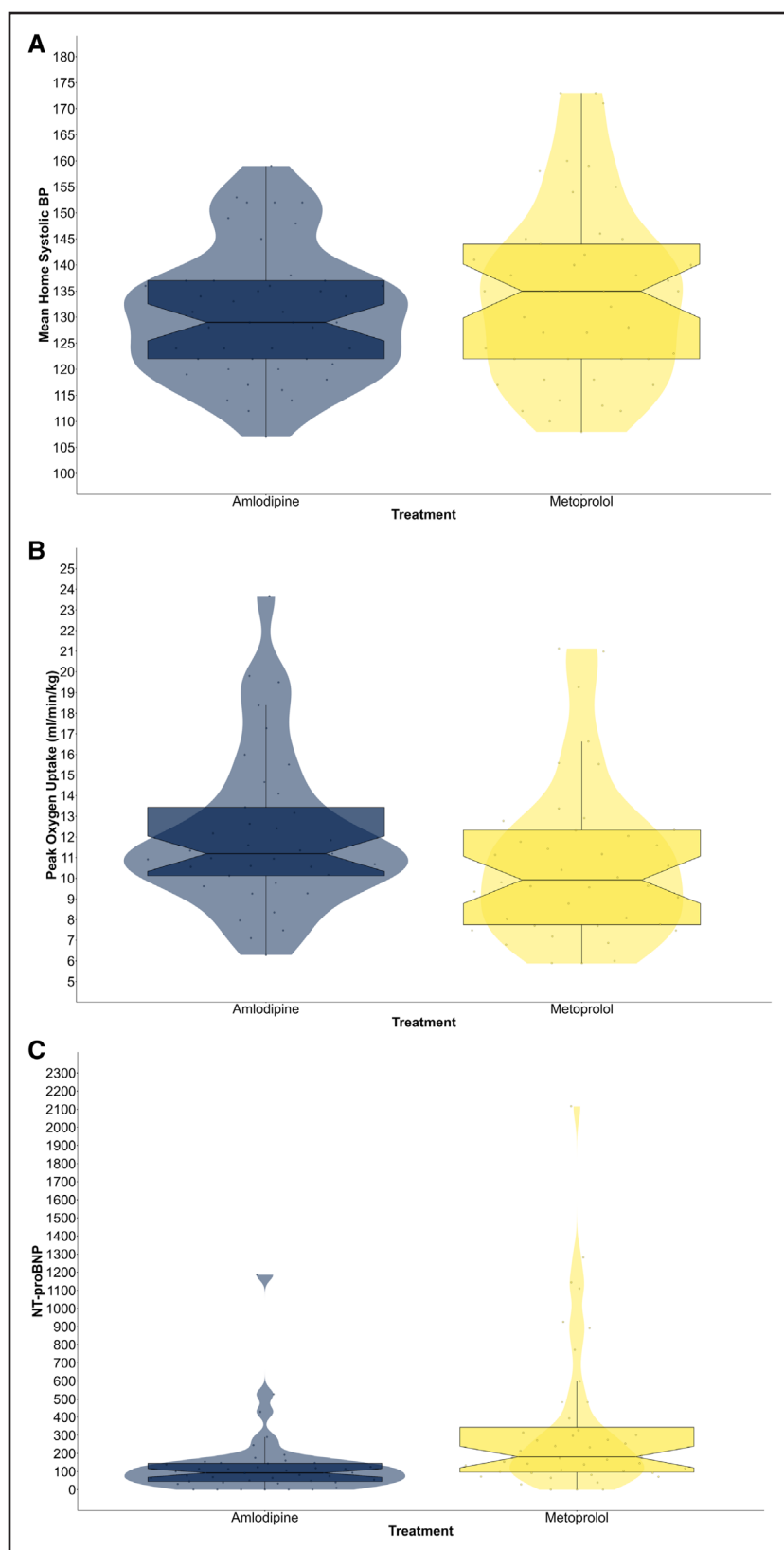


Figure 2. Key outcomes among participants receiving amlodipine vs metoprolol succinate.

A, Shows the distribution of the primary end point, mean home systolic blood pressure (BP), across amlodipine (**left**) vs metoprolol succinate (**right**). The *y* axis represents the range of BPs in mm Hg, and the *x* axis represents the frequency density during each treatment. The inset box shows the median (center line) and 25th–75th percentiles (box lower and upper bounds, respectively). **B**, Shows the distribution of peak VO_2 ; the *y* axis represents the range of oxygen uptake in mL/min per kg. **C**, Shows the distribution of NT-proBNP (N-terminal pro-B-type natriuretic peptide); the *y* axis represents the range of NT-proBNP in pg/mL.

continuation of β -blockers versus placebo-supported withdrawal of β -blockers among 52 individuals with HFpEF who were previously treated with β -blockers for

any indication and who had known chronotropic incompetence. The trial showed a significant improvement in peak VO_2 (2.1 mL/kg per min; $P < 0.001$) and estimated

Table 3. Adjusted Effect of Amlodipine Versus Metoprolol Succinate on the Primary and Selected Secondary End Points Using Mixed-Effects Models

	LS, mean (SE)		Estimated difference between amlodipine and metoprolol (95% CI)	P value
	Amlodipine	Metoprolol		
Primary outcome				
Home systolic BP, mm Hg	131 (3)	134 (3)	−4 (−7 to −1)	0.023
Secondary outcomes				
Home diastolic BP, mm Hg	75 (1)	75 (1)	−0.5 (−2 to 1)	0.41
Office systolic BP, mm Hg	132 (5)	134 (5)	−2 (−14 to 8)	0.61
Office diastolic BP, mm Hg	72 (2)	72 (2)	−0.8 (−5 to 3)	0.69
Peak oxygen uptake, mL/min per kg	13 (1)	11 (1)	1.1 (0.2 to 1.9)	0.018
Physical activity, average METs/d	1.4 (0.1)	1.3 (0.1)	0.1 (0 to 0.1)	0.017
Moderate- to vigorous-intensity physical activity, min/d	59 (7)	49 (7)	10 (0.1 to 20)	0.048
Systemic vasodilatory reserve, %	22 (3)	27 (3)	−5 (−13 to 3)	0.19
KCCQ, score	70 (4)	71 (4)	−1 (−7 to 4)	0.6
Septal E/e′ ratio	13 (1)	14 (1)	−1 (−1 to 0.4)	0.27
LVEF, %	62 (1)	62 (1)	0 (−2 to 2)	0.99
Global longitudinal myocardial strain, %	−19 (1)	−19 (1)	0.5 (−1 to −0.01)	0.048
NT-proBNP, pg/mL	136 (55)	338 (56)	−202 (−296 to −108)	0.0001
Carotid-femoral pulse wave velocity, m/s	10 (1)	10 (1)	1 (−1 to 2)	0.4
Central systolic BP (radial), mm Hg	122 (3)	129 (3)	−7 (−13 to −1)	0.03
Augmentation index (radial), %	140 (5)	151 (5)	−11 (−24 to 1)	0.07

All analyses were adjusted for period and sequence fixed effects and a random intercept for within-subject effects including age, BMI, eGFR, number of BP medications, sex, and COPD status. BMI indicates body mass index; BP, blood pressure; COPD, chronic obstructive pulmonary disease; eGFR, estimated glomerular filtration rate; KCCQ, Kansas City Cardiomyopathy Questionnaire; LS, least squares; LVEF, left ventricular ejection fraction; and NT-proBNP, N-terminal pro-B-type natriuretic peptide.

LV filling pressures (E/e' declined by 1; $P=0.027$) during β -blocker withdrawal versus continuation. Unlike our study, there was no significant difference in NT-proBNP during β -blocker withdrawal versus continuation. Differences in BP across arms were not a focus of the trial and thus were not reported. Moreover, a comparator agent for BP control was not assessed. Interestingly in our study, we did not observe any effect modification by chronotropic incompetence on the effect of amlodipine versus β -blocker therapy on our study end points. Thus, our findings suggest that the benefits of withdrawing β -blockers observed by Palau et al may extend to individuals without chronotropic incompetence, and support the use of amlodipine instead of β -blockers in HFpEF.

There are several potential mechanisms that support our findings. In HFpEF, intact β -adrenergic activity may be a key factor in ensuring exercise tolerance.³⁸ Beyond heart rate, β -blockade may also have lusitropic effects on early relaxation, as well as inotropic effects affecting the cardiac output reserve. Alternatively, the arterial hemodynamic effects of CCBs may reduce LV afterload and improve exercise tolerance in HFpEF,^{16,17} exemplified by the substantial improvement in central BP. Physiological trial data in healthy young adults show that metoprolol increases natriuretic peptides,³⁹ raising the possibility that the between-treatment differences observed here could reflect β -blocker-associated augmentation rather

than solely CCB-associated reduction. Observational evidence suggests no mortality benefit from treatment with CCBs in HFpEF.⁴⁰ However, to our knowledge, there are no prior trials evaluating the potential role of dihydropyridine CCBs in HFpEF. By promoting arterial relaxation and total vascular compliance, dihydropyridine CCBs could reduce the mechanical load on the LV during systole. This may be especially beneficial in HFpEF, a condition in which the heart's ability to increase output during physical activity may be limited, partly due to stiff ventricles and elevated filling pressures.

Participants in our trial represent an obesity- and resistant hypertension-predominant HFpEF phenotype frequently encountered in contemporary practice.⁴¹ While NT-proBNP levels were modest, obesity can attenuate natriuretic peptide concentrations;^{42,43} the NT-proBNP levels in our trial population were comparable to those in prior HFpEF trials conducted in patients with HFpEF and predominant obesity.^{44,45} Entry into the trial required a combination of clinical HF and objective evidence of elevated filling pressures. In addition, the diagnosis was reviewed and confirmed by at least 1 cardiologist with expertise in HFpEF on the study team before enrollment. Participants were receiving a mean of 3 antihypertensive medications at baseline, reflecting the treatment intensity typical of this phenotype. Our findings therefore inform comparative BP management in patients with HFpEF

who are often already treated with multiple agents, rather than initial therapy in untreated hypertension. To contextualize the magnitude of systolic BP difference we observed between amlodipine and metoprolol in this population, the systolic BP effects were comparable to those of spironolactone versus placebo⁴⁶ and sacubitril/valsartan versus placebo⁴⁷ in prior HFpEF trials.

Strengths of our study include that, to our knowledge, this is the first trial to evaluate the comparative effectiveness of specific commonly used antihypertensive agents in patients with hypertension and HFpEF. We also collected detailed information on aerobic capacity, LV function, and arterial load to provide mechanistic insights about the potential roles of these medications in HFpEF. Limitations of the study include the moderate sample size, which limits the ability to draw conclusions from secondary end points and effect modifiers and contributed to imbalance in 2 baseline covariates. However, we note that our paired design maximized the power of the study because each participant served as his or her own control, enhancing our ability to make statistical inferences of the effect of these interventions for a given sample size, relative to parallel-arm designs. Furthermore, the findings were consistent after adjusting for the imbalanced parameters along with several other prespecified covariates, including period and treatment sequence. We did not collect 24-hour ambulatory BP, which has important prognostic implications.⁴⁸ Instead, we collected dense out-of-office BP data using home BPs to facilitate closer safety monitoring and reduce participant burden, while obtaining values that have been demonstrated to be somewhat comparable to daytime ambulatory BPs.^{48,49} Additionally, a lower proportion of our trial participants were on β -blockers at baseline (46%) than in other recent trials (78–82%^{14,15}). This is likely due to the exclusion of individuals with a contraindication to holding their β -blocker, such as persistent atrial fibrillation. This may limit the generalizability of our findings to patients with HFpEF who have other indications for β -blockers in addition to hypertension. Moreover, the absence of a placebo arm limits our ability to determine whether the observed differences reflect beneficial effects of amlodipine, harmful effects of metoprolol succinate, or both. Finally, dose ranges were chosen to reflect common clinical practice and prior hypertension crossover data,²³ but true equipotency may differ in HFpEF; thus, results reflect comparative effectiveness rather than strict dose equivalence.

PERSPECTIVES

Our findings suggest several mechanistic advantages of using dihydropyridine CCBs over β -blockers for treating hypertension in patients with HFpEF. Among individuals who do not have a compelling indication for β -blockade,

dihydropyridine CCBs offer a more favorable therapeutic profile by more effectively lowering home and central systolic BP while also improving aerobic capacity and markers of LV function. These findings challenge the common clinical practice of favoring β -blockers in this population and support reevaluation of antihypertensive strategies in HFpEF. Larger, longer-term trials are warranted to determine whether the observed physiological benefits of CCBs translate into improved clinical outcomes.

ARTICLE INFORMATION

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Supplemental Material

Tables S1–S7
Figures S1–S3
Protocol
References 6,18,22–29,45,50–68

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